

Assignment 4

Machine Learning I

Indian Statistical Institute

Instructor: Swagatam Das

April 5, 2025

1. Suppose that we are given an independent and identically distributed sample of n points $\{y_i\}$ where each point $y_i \sim \mathcal{N}(\mu, 1)$ is distributed according to a normal distribution with mean μ and variance 1. You are going to analyze different estimators of the mean μ .
 - (a) Suppose that we use the estimator $\hat{\mu} = 1$ for the mean of the sample, ignoring the observed data when making our estimate. Give the bias and variance of this estimator $\hat{\mu}$. Explain in a sentence whether this is a good estimator in general, and give an example of when this is a good estimator.
 - (b) Now suppose that we use $\hat{\mu} = y_1$ as an estimator of the mean. That is, we use the first data point in our sample to estimate the mean of the sample. Give the bias and variance of this estimator $\hat{\mu}$. Explain in a sentence or two whether this is a good estimator or not.
 - (c) In the class you have seen the relationship between the MLE estimator and the least squares problem. Sometimes it is useful to use the following estimate

$$\hat{\mu} = \arg \min_{\mu} \sum_{i=1}^n (y_i - \mu)^2 + \lambda \mu^2$$

for the mean, where the parameter $\lambda > 0$ is a known number. The estimator $\hat{\mu}$ is biased, but has lower variance than the sample mean $\bar{\mu} = n^{-1} \sum_i y_i$ which is an unbiased estimator for μ . Give the bias and variance of the estimator $\hat{\mu}$.

2. The k -Means problem aims to minimize the intra-cluster Sum-of-Squares Error (SSE), which can be defined in terms of the intra-cluster variance, or the sum of pairwise distances in each cluster. Prove that the following identity is true:

$$\frac{1}{|C_k|} \sum_{i, i' \in C_k} \sum_{j=1}^p (x_{ij} - x_{i'j})^2 = 2 \sum_{i \in C_k} \sum_{j=1}^p (x_{ij} - \bar{x}_{kj})^2,$$

where $\bar{x}_{kj} = \frac{1}{|C_k|} \sum_{i \in C_k} x_{ij}$ is the j -th feature mean for the mean of cluster C_k .

If two k -Means algorithms are directly implemented from each side of the identity, what would be the difference in algorithmic complexity in the two algorithms and why?

3. *Weighting features by importance:* The k -Means formulation allows each feature to contribute equally to the calculation of the distance metric. We can extend the k -Means formulation to introduce feature weights $w_1, w_2, \dots, w_p$. The feature weights assign importance to features based

on their contribution to the intra-cluster SSE. This allows us to formulate an objective function for a *feature-weighted k-Means*:

$$\min \sum_{j=1}^k \sum_{i \in C_k} \sum_{t=1}^p w_t^\beta (x_{it} - v_{jt})^2,$$

subject to the constraints $\sum_{t=1}^p w_t = 1, 0 \leq w_t \leq 1, \beta > 1$. Assuming the above objective is to be optimized by an Alternating-Optimization algorithm, derive update expressions for each of the variables: the cluster centers v_j , the cluster memberships C_k , and the feature weights w_t (*Hint: How does Fuzzy c-Means update the fuzzy cluster memberships?*)

4. Consider learning a logistic regression model over a set of data points $\{x^{(i)}, y^{(i)}\}, i = 1, 2, \dots, m$. Recall that the log-likelihood for maximum likelihood is given by:

$$LL(\theta) = \sum_{i=1}^m [y^i \log(h_\theta(x^i)) + (1 - y^i) \log(1 - h_\theta(x^i))],$$

where $h_\theta(x^i) = g(\theta^T x^i)$ and $g(z) = \frac{1}{1+e^{-z}}$. Show that $LL(\theta)$ is a concave function of θ .

5. Let $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$ be n d -dimensional samples and A be any no-singular d -by- d matrix. Show that the vector $\mathbf{x}$ that minimizes

$$\sum_{i=1}^n (\mathbf{x}_i - \mathbf{x})^T A^{-1} (\mathbf{x}_i - \mathbf{x})$$

is the sample mean $\mathbf{x}^* = \frac{1}{n} \sum_{k=1}^n \mathbf{x}_k$.

6. A Gaussian mixture model is a family of distributions whose pdf is in the following form:

$$gmm(\mathbf{x}) := \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_k, \Sigma_k),$$

where $\mathcal{N}(\cdot | \boldsymbol{\mu}, \Sigma)$ denotes the Gaussian pdf with mean $\boldsymbol{\mu}$ and covariance Σ , and $\{\pi_1, \dots, \pi_K\}$ are mixture weights satisfying

$$\sum_{k=1}^K \pi_k = 1, \quad \pi_k \geq 0, \quad k \in \{1, \dots, K\}.$$

The Expectation Maximization algorithm for learning a GMM from a set of sample points $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ is outlined below:

- Initialize $\boldsymbol{\mu}_k, \Sigma_k$, and $\pi_k, k \in \{1, \dots, K\}$.
- Repeat the following until convergence:

(a) E-step:

$$\tilde{z}_{ik} \leftarrow \text{Prob}(\mathbf{x}_i \in \text{cluster } k \mid \{(\pi_j, \boldsymbol{\mu}_j, \Sigma_j)\}_{j=1}^K, \mathbf{x}_i),$$

(b) M-step:

$$\{(\pi_k, \boldsymbol{\mu}_k, \Sigma_k)\}_{k=1}^K \leftarrow \arg \max \sum_{i=1}^n \sum_{k=1}^K \tilde{z}_{ik} \left(\log \mathcal{N}(\mathbf{x}_i | \boldsymbol{\mu}_k, \Sigma_k) + \log \pi_k \right)$$

- (a) Consider a simplified GMM where all mixture components share the same covariance matrix, i.e., $\Sigma_k = \Sigma$. Derive the update rule for Σ in the M-step. (Your answer can rely on the value of $\boldsymbol{\mu}_k$ at the current M-step.)

- (b) Consider an even more simplified GMM where all mixture components share a known covariance matrix $\sigma^2 I$, $\sigma^2 > 0$ and I being the identity matrix. Given a set of sample points $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$, we apply the EM algorithm to estimate the means and the mixture weights, and get cluster probabilities for each sample point. Assume the following are true throughout the EM algorithm:

- The mixture weights $\{\pi_1, \dots, \pi_K\}$ are bounded away from zero, i.e, $\exists \epsilon > 0$ such that $\pi_k \geq \epsilon \forall k \in \{1, \dots, K\}$ throughout the iterations.
- Throughout the iterations,

$$\|\mathbf{x}_i - \boldsymbol{\mu}_k\|^2 \neq \|\mathbf{x}_i - \boldsymbol{\mu}_{k'}\|^2 \quad \forall i \in \{1, \dots, n\}, k \neq k'.$$

Show that as $\sigma^2 \rightarrow 0$, the E-step converges to the update rule for γ in the Lloyd's algorithm (see Problem 5 of Homework 3), i.e., the soft assignment becomes hard.

- (c) In this problem you will investigate connections between the EM algorithm and gradient ascent. Consider a GMM where $\Sigma_k = \sigma_k^2 I$, i.e., the covariances are spherical but of different spread. Moreover, suppose the mixture weights π_k 's are known. The log likelihood then is

$$l(\{\boldsymbol{\mu}_k, \sigma_k^2\}_{k=1}^K) = \sum_{i=1}^n \log \left(\sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}_i | \boldsymbol{\mu}_k, \sigma_k^2 I) \right).$$

A maximization algorithm based on gradient ascent is as follows:

- Initialize $\boldsymbol{\mu}_k$ and σ_k^2 , $k \in \{1, \dots, K\}$. Set the iteration counter $t = 1$.
- Repeat the following until convergence:
 - For $k = 1, \dots, K$,

$$\boldsymbol{\mu}_k^{(t+1)} \leftarrow \boldsymbol{\mu}_k^{(t)} + \eta_k^{(t)} \nabla_{\boldsymbol{\mu}_k} l \left(\{\boldsymbol{\mu}_k^{(t)}, (\sigma_k^2)^{(t)}\}_{k=1}^K \right)$$

- For $k = 1, \dots, K$,

$$(\sigma_k^2)^{(t+1)} \leftarrow (\sigma_k^2)^{(t)} + s_k^{(t)} \nabla_{\sigma_k^2} l \left(\{\boldsymbol{\mu}_k^{(t+1)}, (\sigma_k^2)^{(t)}\}_{k=1}^K \right)$$

- Increase the iteration counter $t \leftarrow t + 1$.

Show that with properly chosen step sizes $\eta_k^{(t)}$ and $s_k^{(t)}$, the above gradient ascent algorithm is equivalent to the following modified EM algorithm:

- Initialize $\boldsymbol{\mu}_k$ and σ_k , $k \in \{1, \dots, K\}$. Set the iteration counter $t = 1$.
- Repeat the following until convergence:

- E-step:

$$\tilde{z}_{ik}^{(t+0.5)} \leftarrow \text{Prob}(\mathbf{x}_i \in \text{cluster } k \mid \{(\boldsymbol{\mu}_j^{(t)}, (\sigma_j^2)^{(t)})\}_{j=1}^K, \mathbf{x}_i),$$

- M-step:

$$\{\boldsymbol{\mu}_k^{(t+1)}\}_{k=1}^K \leftarrow \arg \max_{\{\boldsymbol{\mu}_k\}_{k=1}^K} \sum_{i=1}^n \sum_{k=1}^K \tilde{z}_{ik}^{(t+0.5)} \left(\log \mathcal{N}(\mathbf{x}_i | \boldsymbol{\mu}_k, (\sigma_k^2)^{(t)} I) + \log \pi_k \right)$$

- E-step:

$$\tilde{z}_{ik}^{(t+1)} \leftarrow \text{Prob}(\mathbf{x}_i \in \text{cluster } k \mid \{(\boldsymbol{\mu}_j^{(t+1)}, (\sigma_j^2)^{(t)})\}_{j=1}^K, \mathbf{x}_i),$$

- M-step:

$$\{(\sigma_k^2)^{(t+1)}\}_{k=1}^K \leftarrow \arg \max_{\{(\sigma_k^2)\}_{k=1}^K} \sum_{i=1}^n \sum_{k=1}^K \tilde{z}_{ik}^{(t+1)} \left(\log \mathcal{N}(\mathbf{x}_i | \boldsymbol{\mu}_k^{(t+1)}, \sigma_k^2 I) + \log \pi_k \right)$$

v. Increase the iteration counter $t \leftarrow t + 1$.

The main modification is inserting an extra E-step between the M-step for μ_k 's and the M-step for σ_k^2 's. (Hint: the choices of the step sizes should be related to the E-steps.)

7. The linear regression model assumes that the output Y is a linear combination of the input features X plus noise terms ϵ from a given distribution with weights given by β .

We can write this in matrix form by stacking the datapoints as the rows of a matrix X so that x_{ij} is the j -th feature of the i -th datapoint. Then writing Y , β and ϵ as column vectors, we can write the matrix form of the linear regression model as:

$$Y = X\beta + \epsilon$$

where

$$Y = [Y], \epsilon = [\epsilon], \beta = \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_m \end{bmatrix}, \text{ and } X = \begin{bmatrix} X_{11} & X_{12} & \dots & X_{1m} \\ X_{21} & X_{22} & \dots & X_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ X_{n1} & X_{n2} & \dots & X_{nm} \end{bmatrix}$$

Linear regression seeks to find the parameter vector β that provides the best *fit* of the above regression model. One criteria to measure fitness, is to find β that minimizes a given loss function $J(\beta)$.

In class, we have shown that if we assume that $\epsilon_1; \dots; \epsilon_N$ are i.i.d and sampled from the same zero mean Gaussian that is, $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$, then the least square estimate is also the MLE estimate for $P(Y|X; \beta)$.

Robust Linear Regression When we perform least squares linear regression, we make certain idealized assumptions about the vector of errors ϵ , namely, that it is distributed $\mathcal{N}(0, \sigma^2)$. In practice departures from these assumptions occur. Particularly, in cases where the error distribution is heavier tailed than the Normal distribution (i.e. has more probability in tails than the Normal), the least square loss is sensitive to outliers and hence robust regression methods are of interest.

The problem with the least square loss in the existence of outliers (i.e. when the noise term ϵ_i can be arbitrarily large), is that it weights each observation equally in getting parameter estimates. Robust methods, on the other hand, enable the observations to be weighted *unequally*. More specifically, observations that produce large residuals are down-weighted by a robust estimation method.

In this problem, you will assume that $\epsilon_1; \dots; \epsilon_m$ are independent and identically distributed according to a Laplacian distribution (rather than according to $\mathcal{N}(0, \sigma^2)$). That is, each $\epsilon_i \sim \text{Laplace}(0, b) = \frac{1}{2b} \exp(-\frac{|\epsilon_i|}{b})$.

- (a) Provide the loss function $J_{\text{Laplace}}(\beta)$ whose minimization is equivalent to finding the MLE of β under the above noise model.
- (b) Why do you think that the above model provides a more robust *fit* to data compared to the standard model assuming Gaussian distribution of the noise terms?

8. Table 1 describes positive and negative instances of people who were and were not granted credit card. Each row indicates the values observed, and how many times that set of values was observed. For example, (F, Low, +) was observed 10 times, while (F, Low, +) was observed 80 times.

- (a) Compute the sample entropy H for this training data (with logarithms base 2)?

Table 1: Credit Card Application With Two Attributes

Gender	Income	Approved	Counts
F	Low	+	10
F	High	+	95
M	Low	+	5
M	High	+	90
F	Low	-	80
F	High	-	20
M	Low	-	120
M	High	-	30

(b) Calculate the information gains (IG)

$$IG(\text{Gender}) = H(\text{Approved}) - H(\text{Approved}|\text{Gender}) \quad (1)$$

and

$$IG(\text{Income}) = H(\text{Approved}) - H(\text{Approved}|\text{Income}) \quad (2)$$

for this sample of training data?

(c) Draw the decision tree that would be learned by ID3 from this sample of training data.

9. Let $X := \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ be our sample points, and K denote the number of clusters to use. We represent the cluster assignments of the data points by an indicator matrix $\gamma \in \{0, 1\}^{n \times K}$ such that $\gamma_{ik} = 1$ means $\mathbf{x}_i$ belongs to cluster k . We require that each point belongs to exactly one cluster, so $\sum_{k=1}^K \gamma_{ik} = 1$. The K-means method estimates γ by minimizing the following measure of distortion:

$$J(\gamma, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \dots, \boldsymbol{\mu}_K) := \sum_{i=1}^n \sum_{k=1}^K \gamma_{ik} \|\mathbf{x}_i - \boldsymbol{\mu}_k\|^2,$$

where $\|\cdot\|$ denotes the vector 2-norm. The most popular algorithm for minimizing J is due to Lloyd¹ (1957), which alternates between estimating γ and re-computing $\boldsymbol{\mu}_k$'s:

- Initialize $\boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \dots, \boldsymbol{\mu}_K$, and let $C := \{1, \dots, K\}$.
- While the value of J is still decreasing², repeat the following:
 - (a) Determine γ by

$$\gamma_{ik} \leftarrow \begin{cases} 1, & \|\mathbf{x}_i - \boldsymbol{\mu}_k\|^2 \leq \|\mathbf{x}_i - \boldsymbol{\mu}_{k'}\|^2, \forall k' \in C, \\ 0, & \text{otherwise.} \end{cases}$$

Break ties arbitrarily.

- (b) Recompute $\boldsymbol{\mu}_k$ using the updated γ :
For each $k \in C$, if $\sum_{i=1}^n \gamma_{ik} > 0$ set

$$\boldsymbol{\mu}_k \leftarrow \frac{\sum_{i=1}^n \gamma_{ik} \mathbf{x}_i}{\sum_{i=1}^n \gamma_{ik}}.$$

Otherwise, remove k from C .

¹Lloyd, S. P. (1957). "Least square quantization in PCM". Bell Telephone Laboratories Paper.

²This stopping condition is slightly different from the one in Lloyd's algorithm: stop when γ remains the same.

- (a) Show that Lloyd's algorithm stops in a finite number of iterations. (Hint: How many different values can γ take?)
- (b) Let $\bar{\mathbf{x}}$ denote the sample mean. Consider the following three quantities:

$$\begin{aligned} \text{Total variation:} \quad V(X) &:= \frac{\sum_{i=1}^n \|\mathbf{x}_i - \bar{\mathbf{x}}\|^2}{n}. \\ \text{Within-cluster variation:} \quad V_k(X) &:= \frac{\sum_{i=1}^n \gamma_{ik} \|\mathbf{x}_i - \boldsymbol{\mu}_k\|^2}{\sum_{i=1}^n \gamma_{ik}}. \\ \text{Between-cluster variation:} \quad \tilde{V}(X) &:= \sum_{k=1}^K \left(\frac{\sum_{i=1}^n \gamma_{ik}}{n} \right) \|\boldsymbol{\mu}_k - \bar{\mathbf{x}}\|^2. \end{aligned}$$

What is the relation between these three quantities? Based on this relation, show that K-means can be interpreted as minimizing a weighted average of with-in cluster variations while approximately maximizing the between-cluster variation. Note that the relation may contain an extra term that does not appear above.

- (c) Instead of the squared Euclidean distance, we now use the Manhattan distance, denoted by $\|\cdot\|_1$, as the measure of distortion:

$$J_1(\gamma, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \dots, \boldsymbol{\mu}_K) := \sum_{i=1}^n \sum_{k=1}^K \gamma_{ik} \|\mathbf{x}_i - \boldsymbol{\mu}_k\|_1.$$

We minimize J_1 by a variant of the Lloyd's algorithm:

- Initialize $\boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \dots, \boldsymbol{\mu}_K$, and let $C := \{1, \dots, K\}$.
- While the value of J_1 is still decreasing, repeat the following:
 - i. Determine γ by

$$\gamma_{ik} \leftarrow \begin{cases} 1, & \|\mathbf{x}_i - \boldsymbol{\mu}_k\|_1 \leq \|\mathbf{x}_i - \boldsymbol{\mu}_{k'}\|_1, \forall k' \in C, \\ 0, & \text{otherwise.} \end{cases}$$

Break ties arbitrarily.

- ii. Recompute $\boldsymbol{\mu}_k$ using the updated γ :
For each $k \in C$, if $\sum_{i=1}^n \gamma_{ik} > 0$ set

$$\boldsymbol{\mu}_k \leftarrow ?$$

Otherwise, remove k from C .

Fill in the missing update rule for $\boldsymbol{\mu}_k$ such that the algorithm produces a sequence of decreasing objective values unless a local minimum is reached. Note that

- Answers may not be unique.
- The following fact may be useful. Let $\{r_1, r_2, \dots, r_n\}$ be n real numbers. The solution to the minimization problem

$$\min_x \sum_{i=1}^n |x - r_i|$$

is the median of $\{r_1, r_2, \dots, r_n\}$.

- (d) Show that the minimum of J is a non-increasing function of K , thereby conclude that it makes no sense to choose the number of clusters K by minimizing J .

10. In this problem, you will implement the EM algorithm to learn the parameters of a two-class Gaussian mixture model. Recall that a mixture model is a density created by drawing each instance X from one of two possible distributions, $P(X|Y = 0)$ or $P(X|Y = 1)$. Y is a hidden variable over classes that simply indicates the distribution each instance is drawn from. We will assume that $P(Y)$ is a Bernoulli distribution and each $P(X|Y)$ is a 1-dimensional Gaussian with unit variance. The joint density is therefore:

$$P(X = x) = \sum_{y \in \{0,1\}} P(X = x|Y = y) \times P(Y = y)$$

$$P(X = x; \mu, \theta) = \sum_{y \in \{0,1\}} \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{(x - \mu_y)^2}{2}\right\} \times \theta_y$$

The parameters of this model are $\mu = [\mu_0, \mu_1]$ and $\theta = [\theta_0, \theta_1]$, where μ_y is the mean of the Gaussian for class y , and $\theta_y = P(Y = y)$ is the probability that an instance is drawn from class y . (Note that $\theta_0 + \theta_1 = 1$.) We will use EM to estimate these parameters from a data set $\{x^i\}_{i=1}^n$, where $x^i \in \mathbf{R}$.

- (a) Let p_{iy} denote the probability that the i th instance is drawn from class y (i.e., $p_{iy} = P(Y = y|X = x^i)$). During the iteration t , the E-step computes p_{iy} for all i, y using the parameters from the previous iteration, $\mu^{(t-1)}$ and $\theta^{(t-1)}$. Derive an expression for p_{iy} in terms of these parameters.
 - (b) The M-step treats the p_{iy} variables as fractional counts for the unobserved y values and updates μ, θ as if the point (x^i, y) were observed p_{iy} times. Derive an update equation for $\mu^{(t)}$ and $\theta^{(t)}$ in terms of p_{iy} .
11. In this question you should assume that the activation function of a perceptron is the step function. More specifically, this function:
- (a) outputs 0 if the weighted sum of inputs is LESS THAN 0 (not less than or equal).
 - (b) outputs 1 if the weighted sum of inputs is greater than or equal to 0.

Design a perceptron that takes three Boolean inputs (i.e., inputs that are equal to 0 for false, and 1 for true), and outputs: 1 if at least two of the three inputs are true, 0 otherwise. You should NOT worry about what your perceptron does when the input values are not 0 or 1.

12. Is it possible to design a neural network (which could be a single perceptron or a larger network), that satisfies these specs?
- (a) Takes a single input called X , which can be any real number.
 - (b) If $X < 3$, the network outputs 0.
 - (c) If $3 < X < 7$, the network outputs 1.
 - (d) If $X > 7$, the network outputs 0.

We don't care what the network outputs when $X = 3$ or $X = 7$. If your answer is no, explain why not. If your answer is yes, your solution should include a drawing of the network, that shows all edges connecting outputs of one layer to inputs in the next layer, and that also shows the values of all weights (including bias weights) of all perceptrons.